

Chapter 6

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Worksheet for Chapter 6: Critical Thinking with Data

Complete each section of this worksheet individually or in small groups as directed by your teacher. Use a notebook, calculator, or digital tools (like Google Sheets) if needed for analysis or research. Be prepared to discuss your answers in class or present your findings. Let's sharpen your data detective skills and have fun while doing it!

Part 1: Spotting Misleading Data (15 Points)

Objective: Identify tricks in data presentation and understand how graphs and stats can mislead in real-world contexts.

- **Graph Trap Analysis (10 Points)** Look at this scenario: A social media post shows a bar chart claiming, "Teen phone use up 200% in 2 years!" The y-axis starts at 50% instead of 0%, making the increase look massive. a. **What's Misleading? (5 Points):** In 2-3 sentences, explain why this graph might trick viewers.
- b. **How to Fix It? (5 Points):** Suggest one way to make the graph more honest. Write 2 sentences.
- **Media Manipulation Spotter (5 Points)** Think of an ad or news headline you've seen (or imagine one) that used a stat or graph to grab attention (e.g., "90% of teens love this snack!"). In 2-3 sentences, describe it and explain one way it might be misleading (e.g., small sample size).

Part 2: Fact-Checking with Data (20 Points)

Objective: Practice verifying claims using data and evaluate the credibility of sources.

- **Claim Check Challenge (10 Points)** A viral tweet says, "75% of teens skip breakfast and fail exams!" a. **Verification Steps (6 Points):** List 3 specific steps you'd take to fact-check this claim (e.g., search for studies).
 - Step 1: _____
 - Step 2: _____
 - Step 3: _____
- b. **Potential Issues (4 Points):** In 2 sentences, name one reason this claim might be false or exaggerated.
- **Source Reliability Test (10 Points)** You find two sources for a stat on teen stress:
 - Source A: A government health website with a detailed study of 10,000 teens.
 - Source B: A random blog post with no author or data, just opinions. a. **Which to Trust? (5 Points):** Choose the credible source and explain why in 2-3 sentences.
- b. **Why Credibility Matters (5 Points):** In 2 sentences, explain why using credible sources is important for teens like you.
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Part 3: Developing a Data Mindset (25 Points)

Objective: Build critical thinking by questioning assumptions, recognizing biases, and drawing logical conclusions from data.

- **Bias and Assumption Buster (10 Points)** Think of one assumption or bias you or others might have about teens (e.g., "Teens who game a lot are lazy"). a. **Your Assumption/Bias (4 Points):** Write it down. Then, suggest one way to test it with data.
 - Assumption/Bias: _____
 - Test with Data: _____
- **Why Question It? (6 Points):** In 2-3 sentences, explain how questioning this assumption or bias could lead to a fairer or more accurate view.
- **Logical Data Conclusion (15 Points)** Use this mini dataset about 5 students' weekly screen time (hours) and quiz scores:
 - Student 1: 10 hours, Score: 70
 - Student 2: 5 hours, Score: 85
 - Student 3: 15 hours, Score: 60
 - Student 4: 8 hours, Score: 75
 - Student 5: 12 hours, Score: 65
- **Question: "Does more screen time always mean lower quiz scores?"**
- a. **Pattern Spotting (5 Points):** In 2-3 sentences, describe any pattern you see in the data.
- b. **Logical Conclusion (5 Points):** In 2-3 sentences, state a logical conclusion based on this data. Avoid overgeneralizing.
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- c. **Next Step (5 Points):** Suggest one additional piece of data to collect (e.g., sleep hours) to make your conclusion stronger. Explain why in 2 sentences.
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Part 4: Creative Application (10 Points)

Objective: Apply critical thinking to a personal or school-related scenario using data skills.

- **Data Detective Mission (10 Points)** Pick a topic you're curious about in your life or school (e.g., "Does skipping lunch affect my focus?" or "Are school events worth the hype?"). Answer the following:
 - Topic/Question: _____
 - Assumption or Bias You Might Have: _____
 - How to Test with Data (e.g., survey, track): _____
 - One Credible Source You Could Check (if applicable): _____
 - Predict a Finding and Why It Matters (2-3 sentences): _____
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